E80 Poster Template

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E80: Experimental Engineering, SP25

Problem Statement

Water quality control is highly important, both for the safety of people playing in the water and the health of aquatic Flora. Depending on the Temperature and PH regime at the surface of the water (along with some other important conditions for growth), algae blooms may be highly likely. These blooms tend to have strong negative impacts on marine health as they block out sunlight from lower plants, reducing the levels of dissolved oxygen in the water. We attempt to measure relevant parameters to understand the quality of water for supporting plant vs algae growth.

Impact

Our robot aims to measure the quality of water at certain distances from short in order to assess living conditions for aquatic plant life and algae blooms. Algae and plant health are highly dependent on the temperature of water, the PH, and the salinity of the region, and the amount of incoming sunlight (which can be measured by proxy with turbidity levels).

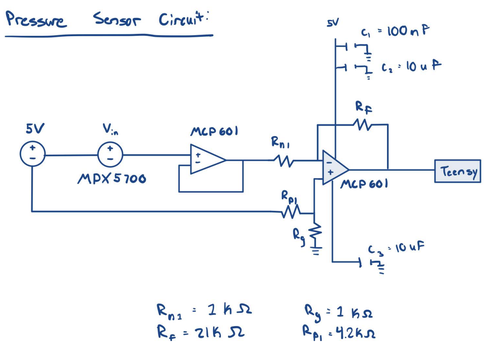
Future Work

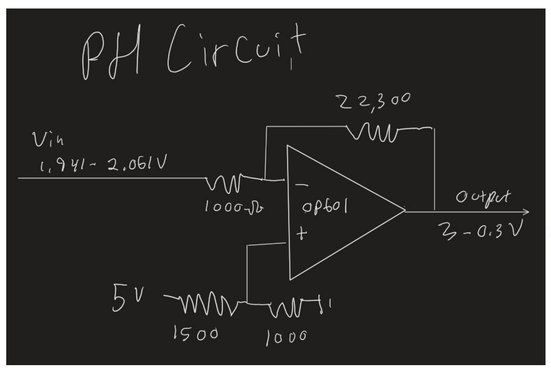
We would like to re-take data with a working PH sensor. Further, we would like to take data at regular intervals going further from the shore, going deeper, and pausing for longer time periods to allow all of our sensors to reach a clear steady state.

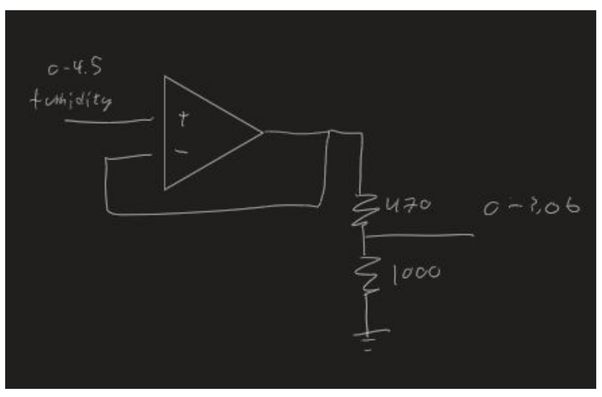
Autonomous Underwater Vehicle (AUV) Features

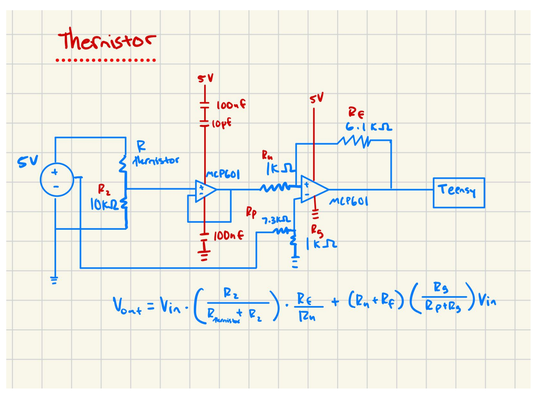
Electrical Schematic & Sensors

Sensors on board: PH, Turbidity, temperature, Salinity, Pressure

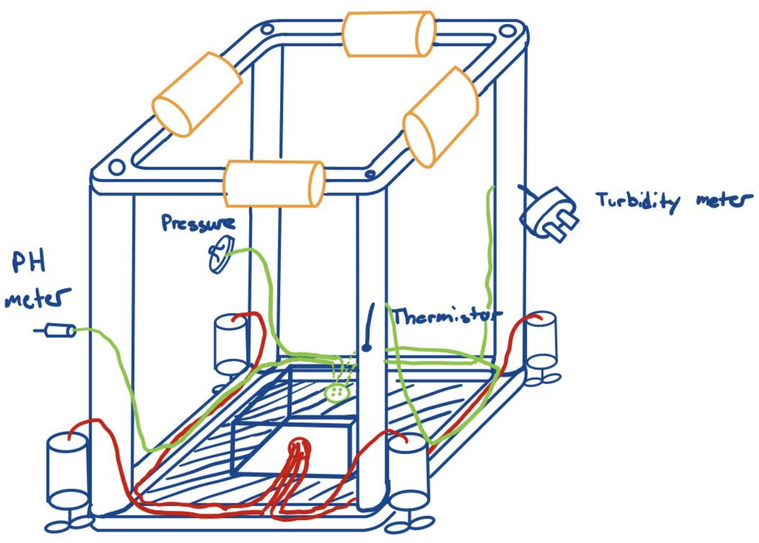


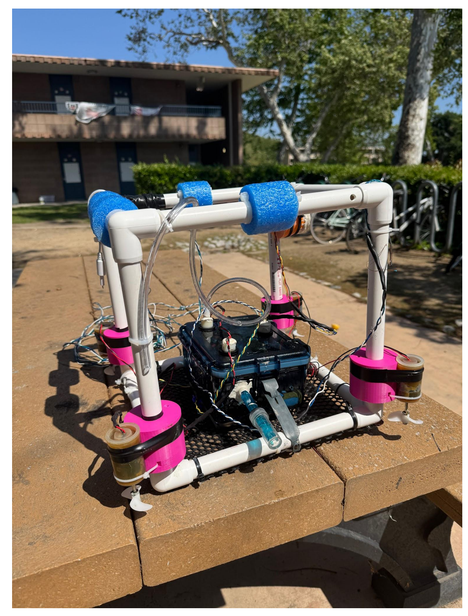






Mechanical Design & Integration

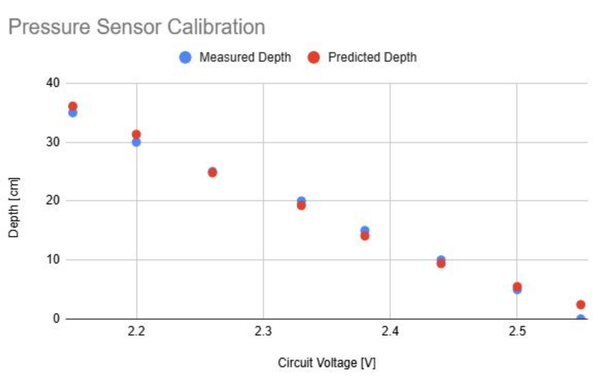




Experimental Data and Results

Sensor Calibration

For each sensor, we used operational amplifiers to yield greater sensitivity within the range of data that we expected to observe at the field station. To calibrate, we compared the output voltage from our amplified circuit to a calibration standard, and found a first order linear regressions.



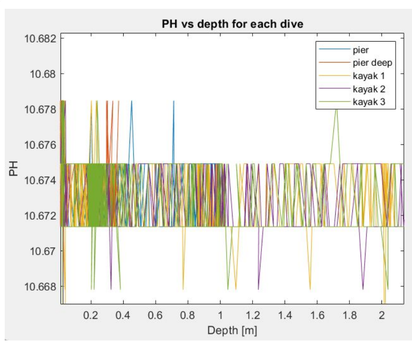
Ground Truth Measurements

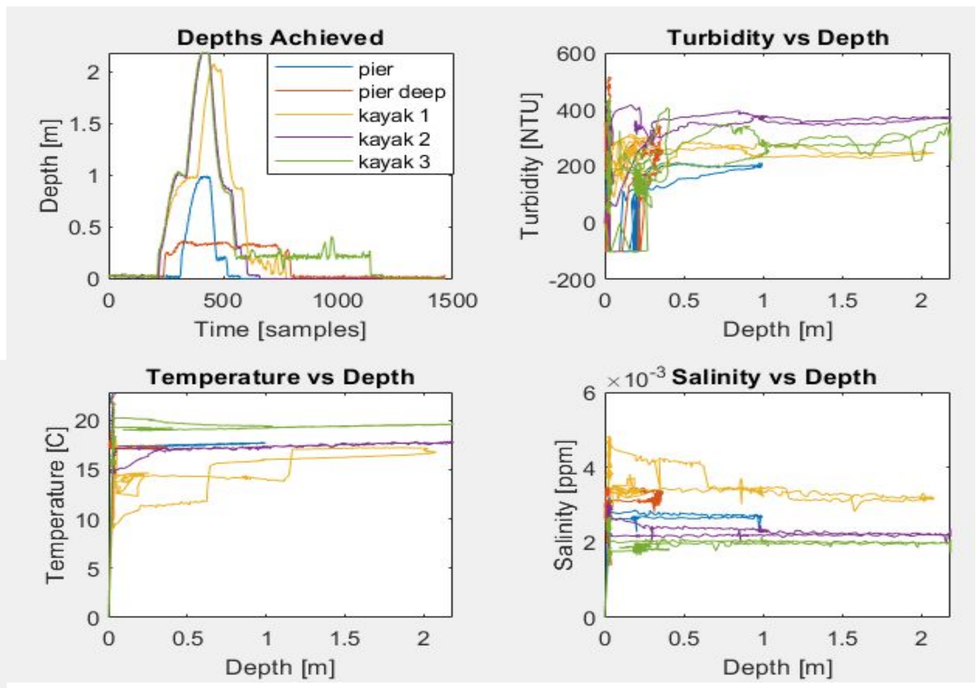
For each sensor, we compared the measurements of temperature and pressure we measured from the air and compared it with the actual values that day, which were 13.9 degrees Celsius and 101.3 kilopascals. For temperature, our graph shows that when the bot was in the air, we measured around 14 degrees celsius which shows we had decent accuracy with the temperature. See sensor calibration for accuracy of the depth measurements.

Results & Conclusions

We see an unexpected trend for turbidity as we get further from the shore. After the sensor's steady state response, we see level curves where the closer to the pier we were, the higher the turbidity. For salinity, we get our expected result with nearly constant ppm levels, only perturbing by extremely small orders of noise. Temperature appears to be warmer further away from the shore, however, which is unexpected. Due to rain, there is potentially a temperature inversion layer explaining the behavior seen in kayak 1.

Our PH sensor was non-operable, and did not yield useful results.

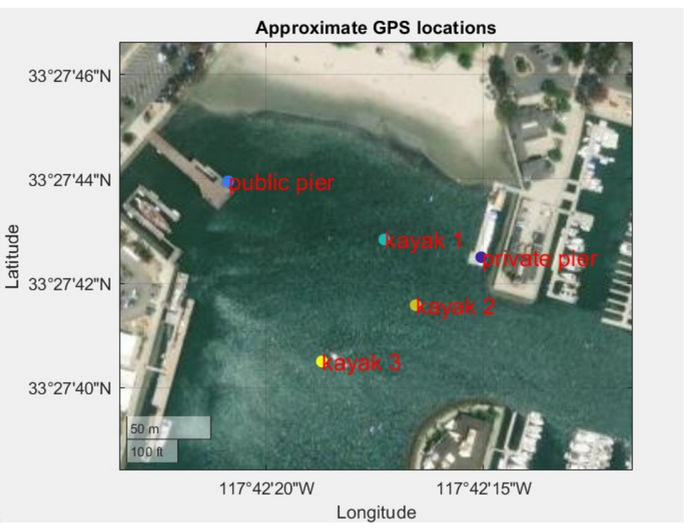




Deployment Details

Location(s): Two piers and 3 spots on the water.

Testing Protocol: Prior to deployment our team team ensured proper buoyancy, tested our motors, and took test data on both piers. We then performed 3 tests in the middle of the water, with the robot dropped from the side of a kayak. Each test, our robot went down to a depth of ~2 meters.



Acknowledgements

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